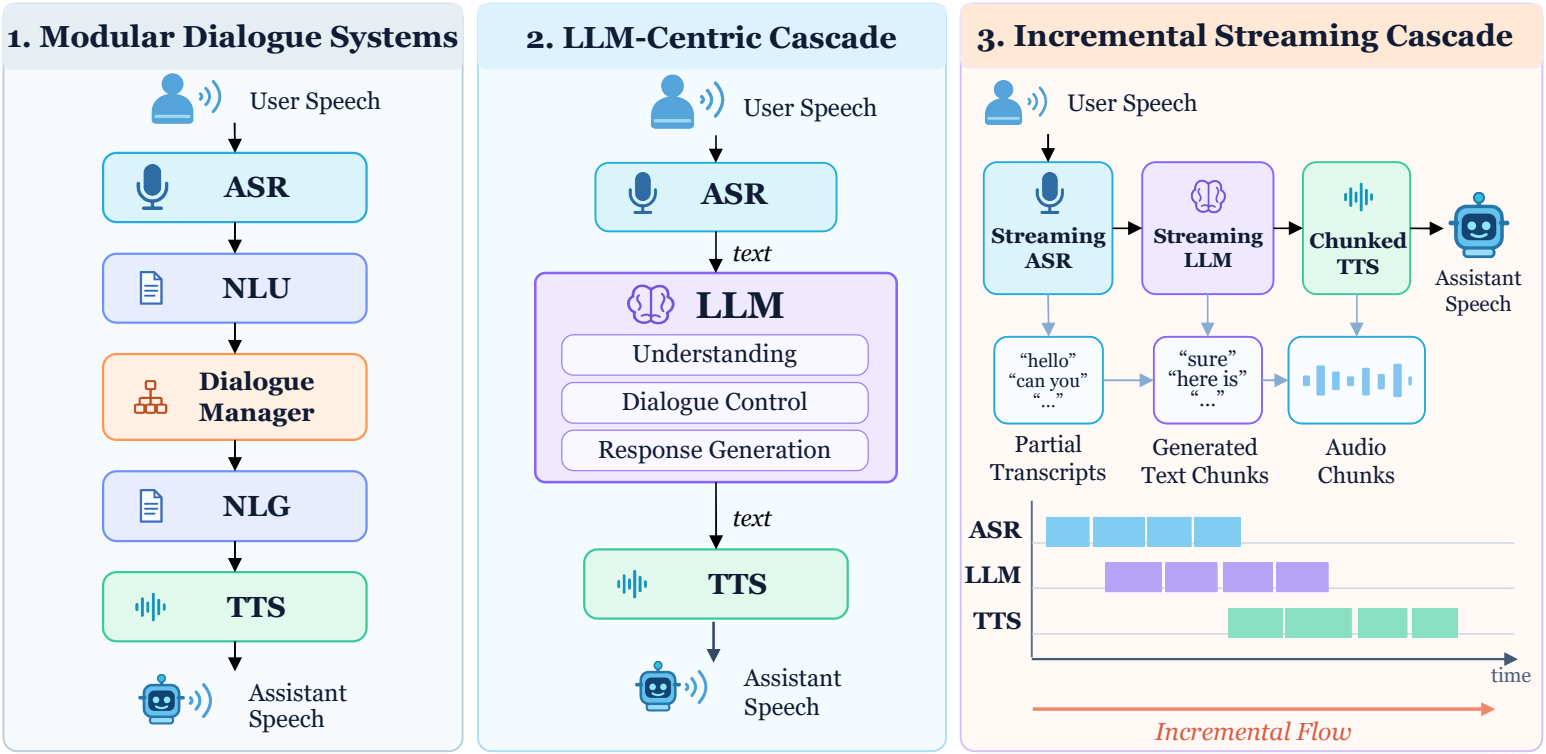

1. Modular Dialogue Systems
User Speech
ASR
NLU
Dialogue Manager
NLG
TTS
Assistant Speech
2. LLM-Centric Cascade
User Speech
ASR
text
LLM
Understanding
Dialogue Control
Response Generation
text
TTS
Assistant Speech
3. Incremental Streaming Cascade
User Speech
Streaming ASR
Streaming LLM
Chunked TTS
Assistant Speech
"hello" "can you" "..."
"sure" "here is" "..."
Partial Transcripts
Generated Text Chunks
Audio Chunks
ASR
LLM
TTS
time
Incremental Flow
Modular Processing
Language Consolidation
Temporal Coordination